

Principles of Economics

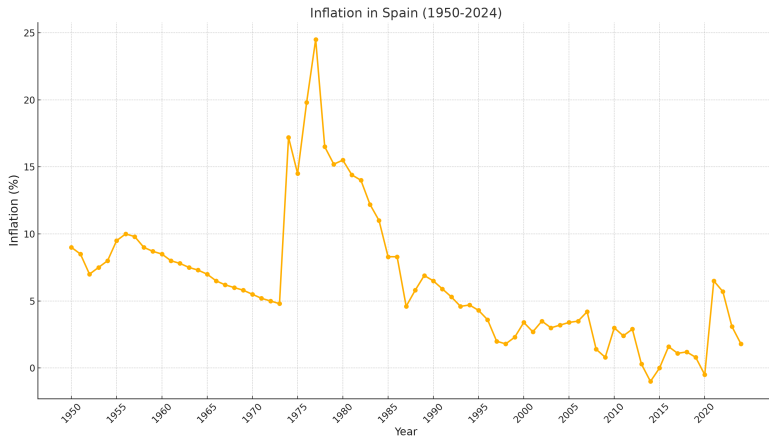
Class 12: Inflation, Unemployment and Monetary Policy

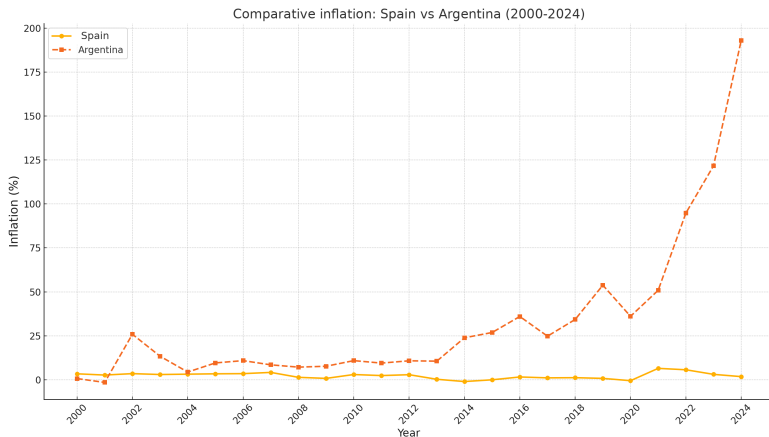
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- ▶ Last class: fiscal policy to smooth fluctuations in **aggregate demand (GDP)**:
 - ▶ Automatic stabilisers, government spending or tax changes [Macro, Unit 5]
 - ▶ Crucial to control output and unemployment fluctuations.
- ▶ **GDP fluctuations** also come with changes in other macro fundamentals: **prices**.
- ▶ Today:
 - ▶ Which **factors affect the price level**?
 - ▶ Causes and consequences of inflation
 - ▶ What is the **optimal level of inflation** and how do central banks **achieve it**?
 - ▶ Inflation-unemployment trade-off [seen in class 10, now in detail]
 - ▶ Expectations: why they matter and how central banks manage them.
 - ▶ How do central banks respond to supply and demand shocks?





Inflation

Consequences of Inflation

Inflation and Unemployment: The Phillips Curve

Inflation + unemployment:
expectations and supply shocks

Monetary Policy: how it works

Monetary policy and stabilisation

Independence of Monetary Policy

Monetary Policy and capacity utilisation

Summary

1. Inflation

- ▶ **Measuring inflation is complicated:**
 - ▶ Turning the **price behaviour of all goods**, sold in different places, **into a single number**
- ▶ A reasonable option: measure the cost of a basket of goods:
 - ▶ **Consumer Price Index (CPI)**: basket of the goods consumed in an economy.
 - ▶ Calculated by the **National Statistics Institute**.
 - ▶ Measures the **cost of living**.
 - ▶ **The GDP deflator**: basket of all goods and services in GDP.
 - ▶ Measures the price of **national output** (ratio of real GDP to nominal GDP);
 - ▶ Lets us **compare GDP across countries and over time**

- ▶ **Inflation** (π) is an increase in the general price level.
- ▶ Measured by the **change in the price index** (details later):

$$\pi_t = \frac{IPC_t - IPC_{t-1}}{IPC_{t-1}}$$

- ▶ Usually relative to the previous year.
- ▶ **Three** connected concepts:
 - ▶ **Zero Inflation:** Constant price level year after year ($\pi_t = 0$);
 - ▶ **Deflation:** General decrease in the price level ($\pi_t < 0$);
 - ▶ **Disinflation:** Decrease in the inflation rate with respect to the previous period ($\pi_t < \pi_{t-1}$);
 - ▶ Prices keep rising, but at a slower pace.
- ▶ Inflation separates nominal from real variables:
 - ▶ **Real interest rate** = Nominal interest rate - Inflation rate [**Fisher equation**]
 - ▶ **Real wage growth** = Nominal wage growth - Inflation rate

- **Example 1:** Suppose a shopping basket with Tomatoes, Oranges and biscuits, and the following prices.
- In this case we assume that **quantities do not change**.

	Yesterday (t-1)			Today (t)		
	Quantity	Price	Spending	Quantity	Price	Spending
Tomatoes	3	€2	3x€2=€6	3	€3	€9
Oranges	10	€1	€10	10	€1.5	€15
Biscuits	2	€3	€6	2	€4.5	€9
Total	-	-	€22	-	-	€33

$$IPC_t = \frac{3 \times 3 + 10 \times 1.5 + 2 \times 4.5}{3 \times 2 + 10 \times 1 + 2 \times 3} = 1.5$$

$$\text{Normalised to year } t-1, IPC_{t-1} = 1$$

- On average, prices rose by 50%

$$\pi_t = \frac{33 - 22}{22} \times 100 = \frac{1.5 - 1}{1} = 50\%$$

- ▶ **Example 2:** Suppose a new shopping basket with Tomatoes, Oranges and biscuits, and the following prices.
- ▶ In this case we assume that **quantities do not change**.

	Yesterday (t-1)			Today (t)		
	Quantity	Price	Spending	Quantity	Price	Spending
Tomatoes	3	€2	3x€2=€6	3	€1	€3
Oranges	10	€1	€10	10	€1	€10
Biscuits	2	€3	€6	2	€10	€20
Total	-	-	€22	-	-	€33

$$IPC_t = \frac{3 \times 1 + 10 \times 1 + 2 \times 10}{3 \times 2 + 10 \times 1 + 2 \times 3} = 1.5$$

Normalised to year $t - 1$, $IPC_{t-1} = 1$

- ▶ On average, prices rose by 50%

$$\pi_t = \frac{33 - 22}{22} \times 100 = \frac{1.5 - 1}{1} = 50\%$$

- ▶ Has the economy grown?

- ▶ **Example 3:** Now with growth.
- ▶ In this case, we assume that **quantities did change**.

	Yesterday (t-1)			Today (t)		
	Quantity	Price	Spending	Quantity	Price	Spending
Tomatoes	3	€2	$3 \times €2 = €6$	6	€3	€18
Oranges	10	€1	€10	20	€1.5	€30
Biscuits	2	€3	€6	4	€4.5	€18
Total	-	-	€22	-	-	€66

$$IPC_t = \frac{3 \times 3 + 10 \times 1.5 + 2 \times 4.5}{3 \times 2 + 10 \times 1 + 2 \times 3} = 1.5$$

Normalised to year $t - 1$, $IPC_{t-1} = 1$, $\pi_t = 50\%$

Here, we **do not change the quantities** in order to focus on prices!

- ▶ PIB_{t-1} (nominal) = 22€ PIB_t (nominal) = 66€ PIB_t (real) = $\frac{66}{1.5} = 44€$
- ▶ What was nominal/real growth?
 - ▶ Growth (nominal) = $\frac{66-22}{22} \times 100 = 200\%$
 - ▶ Growth (real) = $\frac{44-22}{22} \times 100 = 100\%$

Measuring inflation in real life

GDP in Spain (2016 methodology):

- ▶ Design the **representative consumption basket**:
 - ▶ 479 products;
 - ▶ Decide the weight of each good/class in the basket:
 - ▶ Income groups, age, gender, etc...
- ▶ Ensure **geographic representativeness**:
 - ▶ National, regional and provincial level;
 - ▶ 177/8131 municipalities included.
- ▶ Covers **different types of shops**:
 - ▶ 33 000 different shops.
- ▶ Average number of **prices per month**
220000

Groups of goods and services	#
Food and non-alcoholic beverages	170
Alcoholic beverages and tobacco	12
Clothing and footwear	66
Housing and utilities	18
Furnishings and household equipment	57
Health	11
Transport	31
Communications	5
Recreation and culture	43
Education	7
Restaurants and hotels	22
Miscellaneous goods and services	37
Total	479

2. Consequences of Inflation

- ▶ With a fixed nominal income (pension, wages, rents), inflation cuts **purchasing power**:
 - ▶ Fewer goods with the same nominal wage.
- ▶ Inflation affects the **real value of debt**:
 - ▶ Debt is repaid in nominal terms: borrow €1000 today, repay €1000 in a year.
 - ▶ With inflation the real value of debt falls: **good for borrowers, bad for creditors**.
- ▶ **High inflation** makes the **economy work worse**:
 - ▶ It is usually volatile, and unpredictability **raises uncertainty**.
 - ▶ Producers cannot **tell relative price changes from inflation**:
 - ▶ Relative prices matter: do my prices rise because I am doing well or because all prices rise?
 - ▶ **Menu costs**: prices must be updated more often
- ▶ The case of **hyperinflation in Zimbabwe**

- ▶ Would we be better off with **falling prices? (deflation)?**
- ▶ If prices fall, **households postpone consumption** (especially durables), waiting for lower prices:
 - ▶ A **negative aggregate demand shock** that pushes prices lower;
 - ▶ Firms cut output and probably employment.
- ▶ Deflation **increases the real burden of debt**:
 - ▶ May **raise household saving** to reach wealth targets;
 - ▶ Poorer households suffer more (redistributive effects);
- ▶ Overall: **aggregate demand falls and prices drop further**.
 - ▶ **Deflationary spirals**
 - ▶ **Japan** (1995-2015) deflation in 12/21 years; low growth and rising unemployment
- ▶ Consensus: some inflation ($\approx 2\%$) is good, as long as it is stable.

3. Inflation and Unemployment: The Phillips Curve

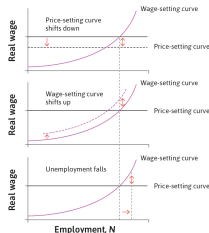
Causes of inflation

Workers, consumers and firms

- ▶ Prices and wages are **set by firms, consumers and workers**.
- ▶ Price level changes respond to factors in **two groups**:

- ▶ **Increase in firms' bargaining power over consumers:**

- ▶ **Less competition** allows firms to raise prices;
- ▶ Shifts the price-setting curve down
- ▶ and a fall in real wages [in the **medium-long run**].



1. Owners' power rises relative to consumers' (e.g. lower competition) – medium to long run

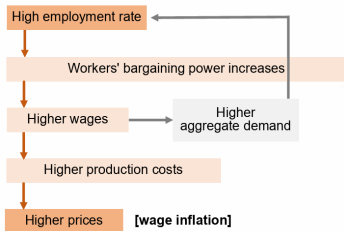
2. Employees' power rises relative to owners' (e.g. stronger unions) – medium to long run

3. Employees' power rises relative to owners' in a business cycle upswing – short to medium run

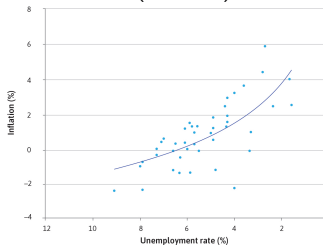
- ▶ **An increase in workers' bargaining power over firms:**

- ▶ **Greater union power** or employment levels;
- ▶ Shifts the wage-setting curve up.
- ▶ and a higher real wage at the same employment level [medium-long run].
- ▶ But movement along the curve [in the **short run**]

- ▶ Rising employment **can generate inflation**:



Wage inflation and unemployment in the UK (1861–1913)



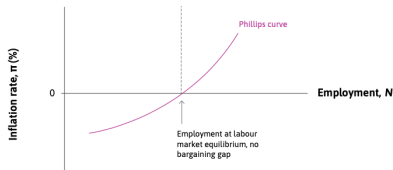
(Unemployment decreases along the horizontal axis, so employment increases)

- ▶ Since there is a **positive relationship between employment and inflation**,
 - ▶ A **business cycle upturn** usually brings **higher inflation**.
 - ▶ Even if **real wages** do not change, **nominal wages and prices** may keep rising → **Inflationary spiral**

The Phillips curve

Introduction

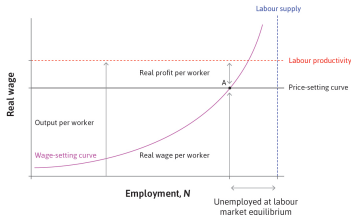
- ▶ The positive relationship between employment and inflation is known as the **Phillips Curve**
- ▶ William Phillips (1914–1975) studied the **relationship between unemployment and inflation**: low unemployment with high inflation, high unemployment with low inflation.
- ▶ It suggests **an employment level with no inflation** (price stability);
- ▶ And that **employment fluctuations bring inflation or deflation**:
 - ▶ You cannot simultaneously have high employment and low inflation!
 - ▶ In fact, **lowering inflation means cutting employment** (raising unemployment)



The Phillips curve

(Un)Employment and inflation

- ▶ **Rationalise this** with the labour market model seen earlier.
- ▶ The **labour market** is in **equilibrium** if:
 - ▶ Given the conditions of the product market (competition, etc.);
 - ▶ The real wage is exactly the one that secures the desired effort.
- ▶ In this scenario:
 - ▶ **Employment, Unemployment and prices (wages and goods) are stable.**
- ▶ This guarantees **zero inflation** [$\pi = 0$]



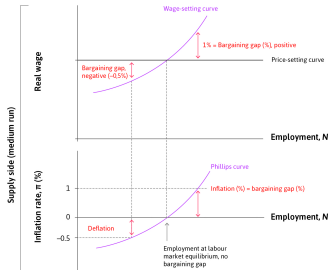
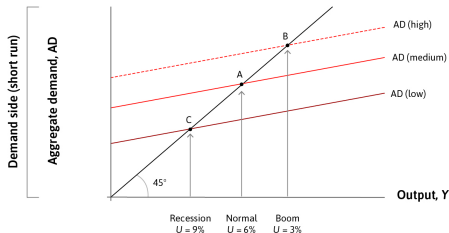
- ▶ Business cycles affect unemployment in the short run. If **unemployment** is:
 - ▶ **Below the equilibrium level:** real wages+profits > total productivity → upward pressure on w/p .
 - ▶ **Above the equilibrium level:** demand for real wages+profits < total productivity → downward pressure on w/p .

- ▶ We summarise the unemployment-inflation link with the bargaining gap:
 - ▶ **Bargaining gap:** gap between the real wage that secures effort and the real wage that leaves firms enough profit.
- ▶ With U^*, E^* the equilibrium levels, **3 cases**:
 - ▶ $U < U^* \iff E > E^*$:
 - ▶ Bargaining gap > 0
 - ▶ $\pi > 0$, prices rise (inflation)
 - ▶ (“too much” employment)
 - ▶ $U = U^* \iff E = E^*$:
 - ▶ Bargaining gap $= 0$
 - ▶ $\pi = 0$, prices are stable
 - ▶ $U > U^* \iff E < E^*$:
 - ▶ Bargaining gap < 0
 - ▶ $\pi < 0$, prices fall (deflation)
 - ▶ (“too little” employment)

The Phillips curve

(Un)Employment and Inflation: Bargaining gap

- ▶ Suppose the **economy is in equilibrium at point A**, with unemployment of 6%;
- ▶ And suppose that:
 - ▶ **AD increases**, leading to a fall in unemployment to 3% [point B]
 - ▶ **AD decreases**, leading to a rise in unemployment to 9% [point C]
- ▶ Changes in employment **create a bargaining gap**.
- ▶ Positive bargaining gap $\rightarrow$ inflation
- ▶ Negative bargaining gap $\rightarrow$ deflation



- ▶ The Phillips Curve points to a relationship between **inflation and unemployment**
- ▶ Policymakers must take this into account when setting the target level.
 - ▶ High inflation, Low unemployment
 - ▶ Low inflation, High unemployment
- ▶ NOTE: CORE has a model of the **optimal (target) inflation choice**:
 - ▶ Policymakers reflect people's **preferences** over inflation [**indifference curves / MRS**]:
 - ▶ Some accept more inflation for less unemployment, and vice versa.
 - ▶ **The Phillips Curve determines the feasible set [MRT]**

4. Inflation + unemployment: expectations and supply shocks

The Phillips curve

Changes over time

- ▶ The relationship is not stable: **the Phillips curve shifts:**

- ▶ In the US, in some periods such as the 70s the relationship is **very steep:**

- ▶ A **small change** in unemployment is enough to change inflation;

- ▶ In other periods such as: 90s, 00s, 10s, the relationship is **very flat:**

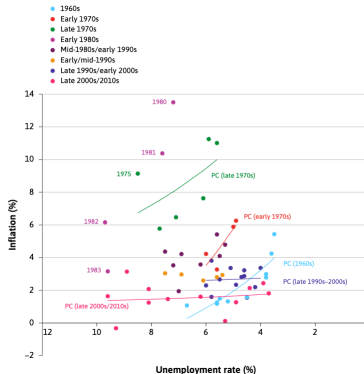
- ▶ A **large change** in unemployment is required to change inflation.

- ▶ Therefore, **there is no permanent trade-off:**

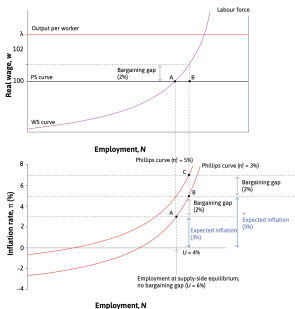
- ▶ The Phillips curve is not a “traditional” feasible set;
- ▶ Only one unemployment rate keeps inflation stable, and it **changes over time.**

- ▶ If a **government keeps unemployment “too low”**, the result is not just higher

The Phillips Curves in the US (1960–2020)



- ▶ **Expectations about future prices shift the Phillips curve:**
Note: Inflation (π) = expected inflation (π^e) + bargaining gap



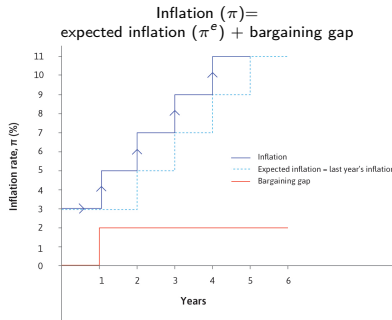
- ▶ Suppose **expected inflation is $\pi^e = 3\%$** :
 - ▶ If the labour market is in equilibrium, **the bargaining gap is zero**:
 - ▶ $\pi = \pi^e = 3\%$ [point A]
 - ▶ A positive demand shock creates an **(unexpected) bargaining gap of 2%**:
 - ▶ $\pi = \pi^e + \text{bargaining gap} = 3\% + 2\% = 5\%$ [point B]
 - ▶ **Negative surprise**: people expected 3% inflation and got 5%!

- ▶ Next period, **expected inflation rises** and the Phillips curve shifts. With $\pi^e = 5\%$ and a 2% gap:
 - ▶ $\pi = \pi^e + \text{bargaining gap} = 5\% + 2\% = 7\%$ [point C]

Changes in inflation

The role of expectations

- ▶ If there is a **positive bargaining gap**,
 - ▶ There will always be **surprise inflation**;
 - ▶ Expectations are revised for next year,
 - ▶ And pushing inflation much higher.

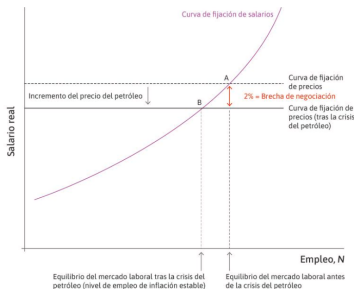


- ▶ On the previous slide, $U = 3\%$ is **below** $U^* = 6\%$: a positive bargaining gap.
 - ▶ As long as this continues, **inflation will keep rising!**
- ▶ The **stable inflation rate** is the unemployment rate that keeps inflation constant.

- ▶ Another **cause of high inflation is a supply shock**:

- ▶ **Supply shock**: unexpected change on the supply side.
E.g. a sharp change in the oil price, war

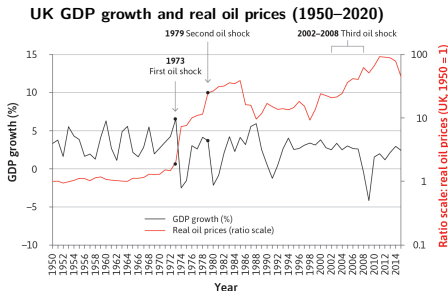
- ▶ Supply shocks shift the **Phillips curve** by affecting the **labour market** equilibrium.



- ▶ With demand constant, **less oil supply**:
Price increases → Downward shift of the price-setting curve → The real wage falls → A positive bargaining gap opens up → **Persistently high inflation**

- ▶ Supply constraints matter, especially when they hit energy costs:
- ▶ **Oil/gas prices** affect the economy:
 - ▶ **Directly:** higher prices for consumers;
 - ▶ **Indirectly:** higher industrial costs (especially transport).

- ▶ In the **70s**, two oil crises led to:
 - ▶ Rising prices (oil and others);
 - ▶ Falling GDP growth;
 - ▶ Spikes in unemployment.



- The inflation spikes in 2021-22 are also connected to **supply disruptions**:

Liberty Street Economics

«Fears That Wages and Input Prices Has Increased in the Post-COVID Period: Main I How Do Natural Disasters Affect U.S. Small Business Owners?»

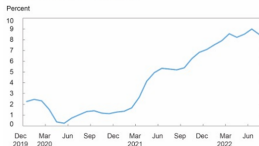
AUGUST 24, 2022

How Much Did Supply Constraints Boost U.S. Inflation?

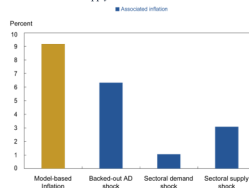
Julian di Giovanni



Inflation has been very elevated



The model calibration shows the quantitative importance of both demand and supply shocks



Source: di Giovanni, Kalemli-Özcan, Silva, and Yıldırım (2022).

Notes: The chart presents a U.S. closed-economy inflation decomposition for a sixty-six sector economy, 2019-21. The first bar shows model-based inflation considering all shocks (demand and supply). The second bar considers the aggregate demand shift only. The third bar uses sectoral demand shocks only. Finally, the fourth bar uses sectoral supply shocks only.

- “Our work shows that US inflation would have been 6% instead of 9% at the end of 2021 without supply bottlenecks.”
- “Put differently, the fiscal stimulus and other aggregate demand factors would not have driven inflation so high without the pandemic-related supply constraints.”
- Read the full article [Here](#).

5. Monetary Policy: how it works

- ▶ **Central banks** run **monetary policy**. Two areas they control:
 - ▶ **Policy interest rate** – the main topic of this class;
 - ▶ **Money supply**;
- ▶ Also, depending on the exchange rate regime, they can intervene in the **exchange rate**:
 - ▶ **Flexible exchange rate**: the currency floats and the market sets its price; the central bank does not intervene. E.g. euro, US dollar, pound sterling.
 - ▶ **Fixed exchange rate**: the currency is pegged to another one. E.g. the Danish krone and several African currencies to the euro; Caribbean and Middle Eastern currencies to the US dollar.
- ▶ The three instruments are linked: policy rate, money supply and exchange rate.

Policy interest rate and monetary policy transmission mechanisms

- ▶ How do **interest rates** affect the economy?

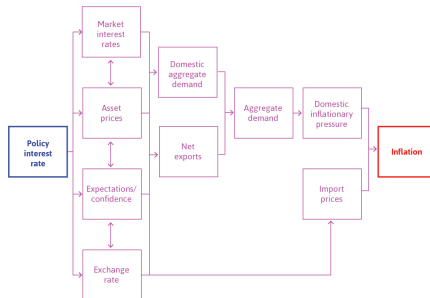
- ▶ **4 transmission channels:**

- ▶ Market interest rates;
- ▶ Asset prices;
- ▶ Expectations/confidence;
- ▶ Exchange rate.

- ▶ Together, they influence:

- ▶ Domestic aggregate demand
- ▶ Net exports;
- ▶ Import prices.

- ▶ Which affects the **price level**.



- ▶ **Bank (or market) interest rates:** the most important channel:
 - ▶ They affect **saving and borrowing, and hence consumption and investment**;
- ▶ **Policy interest rates determine money-market rates that affect market rates** [Class 9]:
 - ▶ Directly: mortgages indexed to Euribor (which is a money-market rate);
 - ▶ Indirectly: the money market sets banks' cost of lending.
- ▶ In reality, the **policy interest rate** is defined, in a way, **backwards**; central banks:
 1. Determine the **desired level of aggregate demand**, based on the labour market and the Phillips curve;
 2. Compute **the real rate** that delivers that aggregate demand (multiplier model);
 3. Compute **the nominal rate** that delivers the right market rate.

- ▶ The **policy interest rate has a domino effect** on market interest rates (loans, mortgage rates) and government debt.
- ▶ Which affects the **prices of all assets**:
 - ▶ When **interest rates fall, the prices of financial assets rise** (bonds, shares, etc.):
 - ▶ Mainly a **substitution effect** between saving and other investments;
 - ▶ Also, **households holding assets get richer** and consume more.
- ▶ Policy interest rates also affect **expectations and confidence**:
 - ▶ **Profit expectations and business confidence** drive firms' investment:
 - ▶ If firms expect low rates, demand and investment rise;
 - ▶ **Household confidence** is also an important determinant of consumption.
- ▶ **Consistency and confidence** are essential for central banks: without them the channel fails.

Policy interest rate and exchange rate

- ▶ **The exchange rate** is the units of domestic currency that can be exchanged for one unit of foreign currency.
- ▶ **The exchange rate between the US dollar (\$-USD) and the euro (€)** can be determined as:

$$\text{Exchange rate } \$/\text{€} = \frac{\text{number of } \$}{1\text{€}}$$

- ▶ An appreciation of the € against the \$ corresponds to a depreciation of the \$ against the €
- ▶ If the exchange rate is 1.15, it means you need \$1.15 to buy €1.
 - ▶ So, a t-shirt costing €20 can be bought with \$23 (20×1.15)
- ▶ **The exchange rate fluctuates.** If the \$/€ rate was 1.15 yesterday and today:
 - ▶ It is 1.00 – there has been a **depreciation of the exchange rate** (the € is worth less today than the \$).
 - ▶ The €20 t-shirt can now be bought with \$20
 - ▶ It is 1.25 – there has been an **appreciation of the exchange rate** (the € is worth more today than the \$)
 - ▶ A €20 t-shirt can now be bought with \$25

- ▶ Rates move the **demand for domestic currency in the FX market**, change its value and with it **net exports** (international competitiveness).
- ▶ Also, dearer imports push prices up and may raise inflation.

The European Central Bank lowers its interest rate.



Demand for European bonds falls.



Demand for euros decreases



The euro depreciates



European exports become cheaper and imports become more expensive.



$(X - M)$ increases



Aggregate demand increases

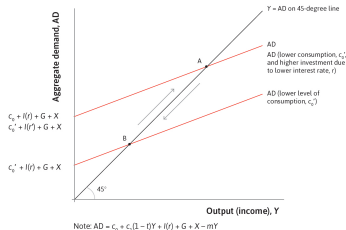
6. Monetary policy and stabilisation

- ▶ Suppose **the economy is initially in equilibrium** [point A], then:
- ▶ **Autonomous consumption falls** [$c_0 > c_0'$]

- ▶ This shifts AD downward;
- ▶ And leads to a new eq. with lower output [point B]

- ▶ To stabilise the economy, the **central bank can:**

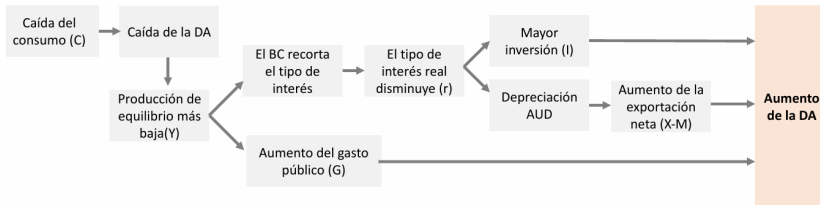
- ▶ **Lower the real interest rate** [$r > r'$];
- ▶ This increases investment [$I(r) > I(r')$];
- ▶ And shifts **AD upward**:
new equilibrium at A!



- ▶ **Monetary policy can be used to stabilise the economy**, as a substitute for or complement to fiscal policy.
- ▶ The central bank runs a **countercyclical policy**: it cuts the rate in the recession, raising AD.

- ▶ From the above, **monetary policy can stabilise the economy:**
 - ▶ Attractive: **no direct budgetary cost.**
- ▶ However, **monetary policy has important limitations:**
 1. Discretionary changes in monetary policy can **erode the credibility of the Central Bank:**
 - ▶ If the rate is high today and falls tomorrow, future investors hold back;
 2. The short-term nominal interest rate (policy rate) cannot fall below zero: **the “zero lower bound” of monetary policy:**
 - ▶ In a crisis, a zero nominal rate may not be low enough;
 - ▶ Alternative: **quantitative easing**, buying financial assets to lower yields and lift investment.
 3. Some countries **do not have monetary policy autonomy:**
 - ▶ In Spain the ECB, not the Bank of Spain, sets the rate.

► Combination of monetary and fiscal policy to stabilise the economy:



► Remember: $AD = C + I + G + X - M$

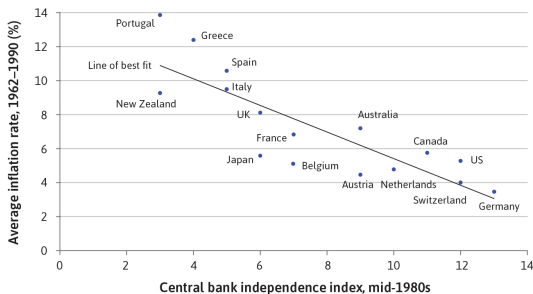
7. Independence of Monetary Policy

- ▶ **Discretionary monetary policy costs the Central Bank credibility;**
- ▶ One alternative is to **define an inflation target and stick to it.**
- ▶ The Central Bank stops depending on **the Government:**
 - ▶ It makes **inflation targets more credible;**
 - ▶ And can avoid inflation spirals:
 - ▶ The rule anchors expectations and prevents **Phillips curve shifts.**

Discretionary policy versus inflation targeting

Central Bank credibility

- ▶ **Central bank independence** has spread since the 1990s:
 - ▶ **Main objective of the CB:** keep inflation around 2%
- ▶ The data: **less independence, higher inflation.**

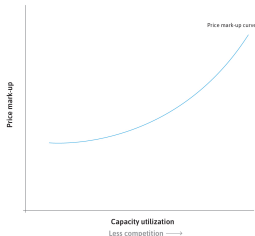


8. Monetary Policy and capacity utilisation

Inflation and low unemployment

Another reason for the relationship

- ▶ So far, **the relationship between (un)employment and inflation appeared through:**
 - ▶ With high employment the cost of job loss falls: firms must raise wages, and prices, to keep profits.
- ▶ There is a second reason: **high capacity utilisation:**
- ▶ **Capacity utilisation is high** if all resources (human and physical) are employed:
 - ▶ **Long run:** firms **invest more** (a bigger factory);
 - ▶ **Short run:** capacity constraints **push prices up**.
- ▶ A wage-price spiral that leads to inflation can be created.



9. Summary

What you take home from each section:

1. **Inflation** → prices are measured with a basket: CPI and GDP deflator; inflation, deflation and disinflation are not the same thing.
2. **Consequences of Inflation** → it erodes purchasing power and the real value of debt; deflation is worse, it feeds spirals.
3. **The Phillips Curve** → employment and inflation move together through the bargaining gap: $U < U^*$ means inflation.
4. **Expectations and supply shocks** → $\pi = \pi^e +$ bargaining gap: the curve shifts, so there is no permanent trade-off.
5. **Monetary Policy: how it works** → the policy rate reaches prices through 4 channels: market rates, asset prices, expectations and the exchange rate.
6. **Monetary policy and stabilisation** → cutting r raises investment and AD, but the zero lower bound and the euro limit it.
7. **Independence of monetary policy** → an inflation target anchors expectations; less independence, more inflation.
8. **Capacity utilisation** → a second route from low unemployment to inflation: capacity constraints push prices up in the short run.

Key concepts: CPI, GDP deflator, deflation, disinflation, Fisher equation, Phillips curve, bargaining gap, expected inflation, supply shock, policy interest rate, transmission channels, zero lower bound, central bank independence, capacity utilisation.